

Anoob prakash

PostDoctoral Research Associate

- ▶ Purdue University
- ▶ Forestry and Natural Resources

Skills

Data Analysis	10 years
R	8+ yrs.
Python	2 yrs.
GIS	6 yrs.
ML	2 yr.
Rmarkdown	8 yrs.
Quarto	4 yr.
Bioinformatics	6 yrs.
UNIX	7 years

Education

2018 - 2025

Ph.D. Plant Biology

University of Vermont

Investigated range-wide genotypic and phenotypic variation in long lived species with molecular and machine learning approaches

2011 - 2014

MSc. Tree Physiology and Breeding

Kerala Agricultural University

Investigated the impact of industrial particulate pollution on trees

Summary

Highly skilled researcher with expertise in ecological and evolutionary genomics, plant biology, climate change and bioinformatics. Experienced in analyzing large-scale genomic data, conducting climate change adaptation studies, and teaching advanced scientific concepts. Adept at both independent and collaborative research in academic and non-academic settings.

Experience

Post Doctoral Research Associate

01/2026 - Present

Department of Forestry and Natural Resources
Purdue University

- Application of machine-learning approaches to investigate climate pre- or mal-adaptation for species with high restoration potential (SHRP)
- Identifying seed sources for regeneration planting in zones of mal-adaptation.

Ph.D. Candidate

2018 - 2025

Department of Plant Biology
University of Vermont

- Led interdisciplinary teams in collecting and managing large, multi-year field datasets; applied genome-wide association studies (GWAS), Bayesian methods, and bioinformatics to identify loci underlying range-wide adaptation.
- Applied molecular and machine learning approaches to investigate the role of introgression in local adaptation, identifying adaptation patterns to inform conservation strategies for fragmented populations.

Graduate Research Intern

2021 & 2022

The Nature Conservancy (TNC)

- Pioneered the integration of genomic data into restoration planning, setting a precedent for collaborative, science-based decision-making between researchers and land managers.
- Guided landscape-scale restoration using genomics, helping plant 58,000 seedlings and establish 38 monitoring plots across 255 acres in partnership with NGOs.

Selected publications

Applications in Plant Science (2024). **Bringing genomics to the field: An integrative approach to seed sourcing for forest restoration.**

Anoob Prakash, Thibaut Capblancq, Kathryn Shallows, David Saville, Deborah Landau, Chad Landress, Tal Jacobs, Stephen Keller.

Ecological Monographs (2023). **Novel genomic offset metrics integrate local adaptation into habitat suitability forecasts and inform assisted migration.**

Susanne Lachmuth, Thibaut Capblancq, **Anoob Prakash**, Stephen R. Keller, Matthew C. Fitzpatrick.

Philosophical Transactions: Biological Sciences (2022). **Genotypic variation and plasticity in climate-adaptive traits after range expansion and fragmentation of red spruce (*Picea rubens* Sarg.).**

Anoob Prakash, Sonia DeYoung, Susanne Lachmuth, Jacquelyne L Adams, Kurt Johnsen, John R Butnor, David M Nelson, Matthew C Fitzpatrick, Stephen R Keller.

Fellowship

2023 - 2024

GradCAMP Climate Scholar

USDA and West Virginia State University

2018 - 2023

NSF Research Trainee under Quantitative and Evolutionary STEM Training (QuEST)

University of Vermont

Tools

- ▶ Git
- ▶ RStudio
- ▶ VS Code
- ▶ Terminal

Contact

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- 📞 +1 802 777 6768
- ✉ anoobvinu@gmail.com
- 🏠 website
- 🐙 github
- 🎓 scholar

Projects

See my github profile for a comprehensive list of open source projects.

R packages

Creator and Maintainer

edgebundle, graphlayouts, levelnet, netrankr, netUtils, networkdata, PSAWR, roughnet, roughsf, rtot, Rtumblr, signnet, snahelper, webbotparseR, webtrackR

R packages

Contributor

backbone, ggraph, multigraphr, netropy, rang, rgraph6

Quarto extensions

Creator and Maintainer

Academicons (shortcodes), Blackboard (revealjs theme), sketchy (html theme), share buttons (filter), nutshell (filter), quartocities (website template), quartocv (cv templates)

soccerverse.com (football analytics website)

Creator and Maintainer

Manual and automated gathering of football results around the world, Harmonizing data (e.g. club names and managers), Implementation of ranking algorithms, Predicting of league tables and matchforecasting, Uses R, JavaScript, HTML, and CSS, around 1000 visitors/month